

Wide Local Excision for Primary Cutaneous Melanoma

The standard therapy for primary cutaneous melanoma is wide local excision (WLE). The goal of WLE is to prevent local recurrence by achieving complete tumor removal with histopathologically confirmed tumor-free margins.

True local recurrence due to inadequate excision, although occurring in a small subset of patients, can have lethal consequences. Such recurrences are considered potentially preventable when narrow margins are used, particularly in thin melanomas (see figure below).

▲ FIGURE 124-21 Potentially preventable deep local recurrence at both ends of a previous excision scar with satellite lesions, occurring approximately 4 years after narrow excision of a thin melanoma with 0.5-cm margins.

Recommended Clinical Margins

Current guidelines recommend the following clinical margins around the residual lesion or biopsy scar:

- **Melanoma in situ (non-LM pattern):** 0.5 to 1 cm
- **Melanoma <1 mm Breslow depth:** 1 cm
- **Melanoma 1–2 mm Breslow depth:** 1 to 2 cm (as anatomically feasible)
- **Melanoma >2 mm Breslow depth:** 2 cm

In all cases, pathologic confirmation of clear peripheral margins is required.

Historically, surgical margins of 3 to 5 cm were standard. However, multiple large randomized controlled trials have shown no difference in survival or local recurrence rates between narrow (1–2 cm) and wide (3–5 cm) margins.[106–112]

Individualized Surgical Planning

Each patient should be evaluated individually, considering: - Current surgical guidelines

- Anatomic site (e.g., proximity to vital structures)
- Feasibility of primary closure versus need for skin graft or flap reconstruction
- Presence or absence of adverse prognostic factors from microstaging

Special sites—such as the digits, soles, ears, vagina, or anus—require tailored surgical and functional considerations. Despite these challenges, the fundamental oncologic principle remains: **tumor clearance first, reconstruction second.**

For subungual melanoma, **partial or complete amputation of the digit** remains the preferred surgical approach.

Lentigo Maligna and Lentigo Maligna Melanoma

Lentigo maligna (LM) and lentigo maligna melanoma (LMM) are subtypes of melanoma known for extensive **subclinical peripheral spread**, often obscured by background photodamage, making clinical margin delineation difficult.

Histologic confirmation of negative margins is especially critical in these cases due to high rates of microscopic extension beyond visible borders. There is also an increased risk of a macular invasive desmoplastic component in LMM.

Margin Requirements for LM and LMM

Standard margins are frequently inadequate: - **0.5 cm for LM** and **1 cm for LMM** <1 mm clear only about **50%** of lesions.

Studies indicate significantly wider margins are often needed: - **LM:**

- Average margin to clear: 0.8 cm
- 1.5 cm margin clears **96%** of lesions
- **LMM:**
- Mean margin to clear: 1.1 cm
- 2.6 cm margin clears **95%** of lesions[113]

Another study found: - 97% of melanomas **<1.5 cm** in diameter required a **1.5-cm** margin for complete excision
- 95% of melanomas **>3 cm** in diameter required a **2.5-cm** margin[114]

Recurrence Rates

Recurrence rates following standard excision for lentigo maligna range from **8% to 20%**, largely due to positive or narrow margins.

Given these challenges, techniques such as **Mohs micrographic surgery** or **delayed excision with margin assessment** are often considered for LM and LMM, especially in cosmetically or functionally sensitive areas.